

QUESTIONS

Q1

Student Management System
[BL3 – Apply]
10 marks

Q2

Library Book Management [BL3 – Apply]
10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q4 Shape Area Calculator using Polymorphism [BL3 – Apply]

10 marks

Create a Shape base class with an overridable calculateArea() method. Apply this to:

- Circle — area using radius
- Rectangle — area using length and breadth
- Triangle — area using base and height

Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.

OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
class Shape {  
    public double calculateArea() {  
        return 0;  
    }  
}
```

```
class Circle extends Shape {  
    double radius;
```

```
    Circle(double radius) {  
        this.radius = radius;  
    }  
}
```

Q3 ✓
Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4 ✓
Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5 ✓
5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6 ✓
6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7 ✓
7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8 ✓
8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9 ✓
9. University Staff Hierarchy

Your Answer

```
1
  @Override
  public double calculateArea() {
    return Math.PI * radius * radius;
  }
}

class Rectangle extends Shape {
  double length;
  double breadth;

  Rectangle(double length, double breadth) {
    this.length = length;
    this.breadth = breadth;
  }

  @Override
  public double calculateArea() {
    return length * breadth;
  }
}

class Triangle extends Shape {
  double base;
```

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

Q10

10. Smart Home Device

double height;

```
Triangle(double base, double height) {  
    this.base = base;  
    this.height = height;  
}
```

@Override

```
public double calculateArea() {  
    return 0.5 * base * height;  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        Shape shape1 = new Circle(5);  
        Shape shape2 = new Rectangle(4, 6);  
        Shape shape3 = new Triangle(8, 10);  
  
        System.out.println("Circle Area: " + shape1.calculateArea());  
        System.out.println("Rectangle Area: " + shape2.calculateArea());  
        System.out.println("Triangle Area: " + shape3.calculateArea());  
    }  
}
```

 Save Answer